

Time Tracker - Software Requirements Specification (SRS)

1. Introduction

1.1 Purpose

This document specifies the software requirements for Time Tracker, a proposed Progressive Web Application (PWA) designed for personal time management and productivity tracking. The application aims to help users track their time usage, set daily goals, and improve time management through various timing tools and analytics.

1.2 Document Conventions

- Requirements are categorized by priority: High (H), Medium (M), Low (L)
- Functional requirements are numbered as F-001, F-002, etc.
- Non-functional requirements are numbered as NF-001, NF-002, etc.

1.3 Intended Audience and Reading Suggestions

- Client/Stakeholders: Focus on Sections 1.4, 2.2, 2.3 for understanding project scope and features
- Development Team: Focus on Sections 3, 4, and 5 for technical requirements
- Testing Team: Focus on Section 3 for test case development
- Project Managers: Review entire document for project planning

1.4 Project Scope

Time Tracker is proposed as a web-based productivity application that will allow users to:

- Track time using countdown timers and stopwatches
- Set and monitor daily time-based goals

- View progress through visual dashboards
- Maintain historical records of timing sessions
- Use the application offline as a Progressive Web App

Exclusions: Social features, team collaboration, cloud synchronization, and advanced reporting are out of scope for initial release.

1.5 References

- PWA Standards Documentation
- Web Storage API Specifications
- Responsive Web Design Guidelines
- Accessibility Standards (WCAG 2.1)

2. Overall Description

2.1 Product Perspective

Time Tracker will be developed as a standalone Progressive Web Application (PWA) using modern web technologies (HTML5, CSS3, JavaScript). It will operate independently without backend dependencies for core functionality, storing all data locally in the browser.

2.2 Product Features (Planned)

1. Dashboard: Overview of current time, daily totals, and goal progress
2. Countdown Timer: Configurable timer with progress visualization
3. Stopwatch: Lap-based timing with session recording
4. Goal Management: Daily time goal creation and tracking
5. History & Analytics: Session history with filtering capabilities
6. Theme Management: Light/dark mode support
7. PWA Features: Offline functionality and installability

2.3 User Classes and Characteristics

- Students: Need to track study sessions and breaks
- Professionals: Want to monitor work hours and productivity
- Freelancers: Require time tracking for billing and project management
- General Users: Seeking better time management habits

2.4 Operating Environment

- Browsers: Modern browsers with ES6+ support (Chrome 80+, Firefox 75+, Safari 13+)
- Platforms: Windows, macOS, iOS, Android devices
- Screen Sizes: Responsive design supporting mobile, tablet, and desktop
- Storage: Browser localStorage for data persistence

2.5 Design and Implementation Constraints

- Must function entirely client-side for initial release
- Must follow PWA standards for offline capability
- Must use responsive design principles
- Must maintain data privacy by storing all data locally
- Should have a clean, intuitive user interface

2.6 User Documentation

- In-app tooltips and guidance
- Help section explaining features
- Responsive design for mobile accessibility

2.7 Assumptions and Dependencies

- Users have modern browsers with JavaScript enabled
- localStorage API is available and functional
- Service Workers are supported for PWA features
- Users understand basic time management concepts

3. System Features (Proposed)

3.1 Dashboard Module

Priority: High

- Description: Main landing page showing overview statistics and quick access
- Functional Requirements:
 - F-001: System shall display current time updating in real-time
 - F-002: System shall calculate and show total time tracked for current day
 - F-003: System shall display active session status when applicable
 - F-004: System shall provide quick-start buttons for common timer durations
 - F-005: System shall show visual progress indicators for daily goals

3.2 Timer Module

Priority: High

- Description: Configurable countdown timer with visual feedback
- Functional Requirements:
 - F-006: System shall allow users to set timer duration in minutes and seconds
 - F-007: System shall display visual progress indicator (progress ring)
 - F-008: System shall provide preset timer durations (5, 10, 25 minutes)
 - F-009: System shall allow linking timers to specific goals
 - F-010: System shall notify users when timer completes
 - F-011: System shall calculate and display estimated end time

3.3 Stopwatch Module

Priority: High

- Description: Traditional stopwatch with lap functionality
- Functional Requirements:
 - F-012: System shall track elapsed time with precision
 - F-013: System shall support lap recording during timing

- F-014: System shall allow saving sessions with optional notes
- F-015: System shall display lap times in a table format
- F-016: System shall enable/disable controls based on stopwatch state

3.4 Goal Management Module

Priority: Medium

- Description: Creation and tracking of daily time-based goals
- Functional Requirements:
 - F-017: System shall allow creation of goals with name and daily target
 - F-018: System shall automatically track progress toward goals
 - F-019: System shall reset goal progress daily
 - F-020: System shall provide visual indicators of goal completion status
 - F-021: System shall allow deletion/modification of existing goals

3.5 History Module

Priority: Medium

- Description: Record and analysis of past timing sessions
- Functional Requirements:
 - F-022: System shall store all completed sessions with timestamps
 - F-023: System shall allow filtering of history by date, type, and goal
 - F-024: System shall provide summary statistics for filtered periods
 - F-025: System shall allow users to clear history (with confirmation)
 - F-026: System shall persist history data across browser sessions

3.6 Settings Module

Priority: Low

- Description: Application preferences and configuration
- Functional Requirements:
 - F-027: System shall support light/dark theme toggling

- F-028: System shall remember user's theme preference
- F-029: System shall provide basic sound control options
- F-030: System shall display version information

3.7 PWA Features

Priority: Medium

- Description: Progressive Web App capabilities
- Functional Requirements:
 - F-031: System shall be installable as a standalone app
 - F-032: System shall function offline for core features
 - F-033: System shall cache essential assets for offline use
 - F-034: System shall provide appropriate app icons and splash screens

4. External Interface Requirements

4.1 User Interfaces

- Dashboard: Clean layout with statistics cards and quick actions
- Timer: Circular progress display with controls below
- Stopwatch: Digital display with lap table
- Goals: Form-based creation with progress visualization
- History: Tabular data display with filtering options
- Navigation: Top navigation bar for desktop, bottom bar for mobile
- Consistency: Uniform design language throughout application

4.2 Hardware Interfaces

- Input: Support for mouse, keyboard, and touch inputs
- Output: Screen display with optional audio notifications
- Storage: Browser localStorage for data persistence
- Connectivity: Network status detection for PWA features

4.3 Software Interfaces

- Browser APIs: localStorage, Service Workers, Web App Manifest
- JavaScript: ES6+ features for modern functionality
- CSS: Responsive design frameworks (to be selected)

4.4 Communications Interfaces

- Initial Load: HTTP/HTTPS for loading application assets
- Updates: Manual refresh for application updates
- Data: No external API dependencies planned for initial release

5. Other Nonfunctional Requirements

5.1 Performance Requirements

- NF-001: Application shall load within 3 seconds on typical connections
- NF-002: User interface shall respond to interactions within 200ms
- NF-003: Timer accuracy shall be within ± 100 ms over 60-minute periods
- NF-004: Dashboard updates shall occur within 100ms of time changes

5.2 Safety Requirements

- NF-005: Application shall not collect personal user data
- NF-006: All user data shall remain on local device
- NF-007: Notifications shall be non-intrusive and user-configurable

5.3 Security Requirements

- NF-008: No authentication required for local-only application
- NF-009: No external data transmission for user data
- NF-010: Data clearance shall require explicit user action

5.4 Software Quality Attributes

- Usability: Intuitive interface requiring minimal training
- Reliability: Core timing functions must work consistently
- Maintainability: Modular code structure for easy updates
- Portability: Cross-browser compatibility
- Accessibility: Basic accessibility features (keyboard navigation, color contrast)

5.5 Business Rules

- BR-001: Goal progress resets at midnight local time
- BR-002: Session duration calculations use system time
- BR-003: Timer presets follow common productivity techniques (Pomodoro, etc.)

6. Other Requirements

6.1 Installation Requirements

- Application shall be accessible via standard web browsers
- PWA installation shall be optional
- No special permissions required beyond standard browser capabilities

6.2 Data Migration

- No data migration requirements for initial release
- All data stored locally with browser storage

6.3 Legal and Compliance

- Application shall comply with general web standards
- Privacy policy shall state data storage practices
- Open source attribution if using third-party libraries

6.4 Documentation Requirements

- In-app help and guidance
- Basic user manual explaining features
- Developer documentation for future maintenance

Appendix A: Glossary

- PWA: Progressive Web Application
- localStorage: Browser-based key-value storage
- Service Worker: Background script for offline functionality
- Pomodoro: Time management method using 25-minute intervals
- Session: Single timing instance (timer or stopwatch run)
- Lap: Individual segment within a stopwatch session

Appendix B: Future Considerations

- Cloud synchronization across devices
- Team collaboration features
- Advanced analytics and reporting
- Integration with calendar applications
- Export functionality for session data

Appendix C: Open Issues

- Specific color scheme and branding guidelines
- Sound selection for notifications
- Performance testing on low-end devices
- Browser compatibility matrix finalization

Document Version: 1.0

Status: Draft - For Planning & Approval

Project Phase: Pre-Development

Assumptions: Product has not been developed yet; this SRS serves as planning document

Next Steps: Client approval, technical design, development planning